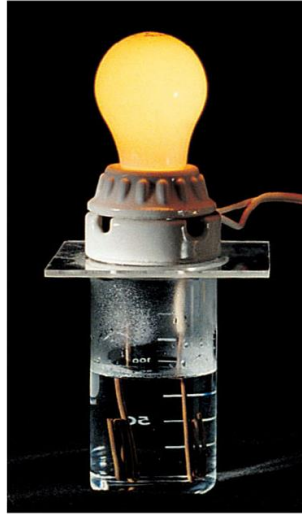


CH 11- Solutions & Colloids Pt2

- Introduction
- 11.1 The Dissolution Process
- 11.2 Electrolytes
- 11.3 Solubility
- 11.4 Colligative Properties
- ~~11.5 Colloids~~



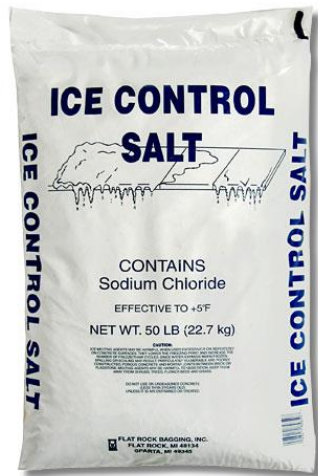
strong electrolyte

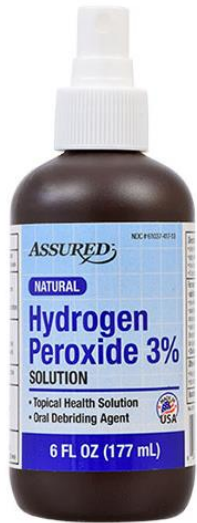


weak electrolyte



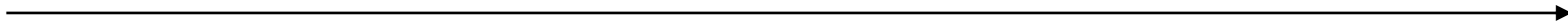
nonelectrolyte





**6% H_2O_2 for
hair bleach**

PT. Mulia Agung
Chemindo



Solution Concentration

- 1. Mole Fraction (χ)

moles of solute $\div$ total moles of solution

So, $\chi_A = 0.5$ means half the molecules are A.

- 2. Molarity ($M = \text{mol}\cdot\text{L}^{-1}$)

moles of solute $\div$ liters of solution

- 3. Molality ($m = \text{mol}\cdot\text{kg}^{-1}$)

moles of solute $\div$ mass of solvent (kg)

Not affected by thermal expansion/contraction of solvent.

Concentration Term	Ratio
Molarity (mol/L) (c) or (M)	$\frac{\text{amount (mol) of solute}}{\text{volume (L) of solution}}$
Molality (m)	$\frac{\text{amount (mol) of solute}}{\text{mass (kg) of solvent}}$
Mole fraction (X)	$\frac{\text{amount (mol) of solute}}{\text{amount (mol) of solute} + \text{amount (mol) of solvent}}$

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- To convert a term based on amount (mol) to one based on mass, you need the *molar mass*.
- To convert a term based on mass to one based on volume, you need the solution density.
- Molality involves quantity of *solvent*, whereas molarity involves quantity of *solution*.

Solutions of Electrolytes

- Electrolyte solutions conduct electricity.
- Many important physiological solutions are electrolytes.

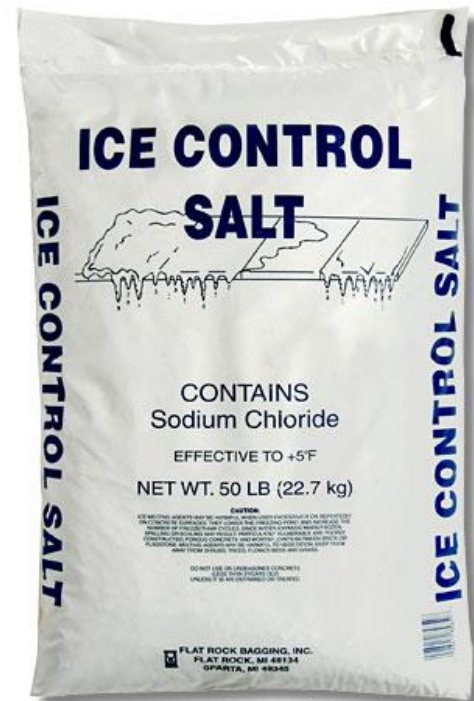


Solutions of Electrolytes, cont'd

What makes a substance an electrolyte?

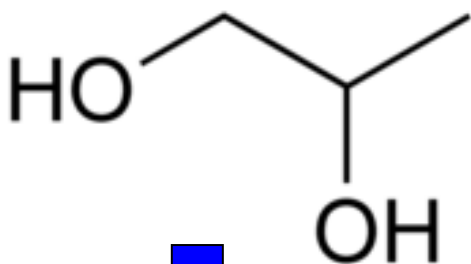


non-electrolyte



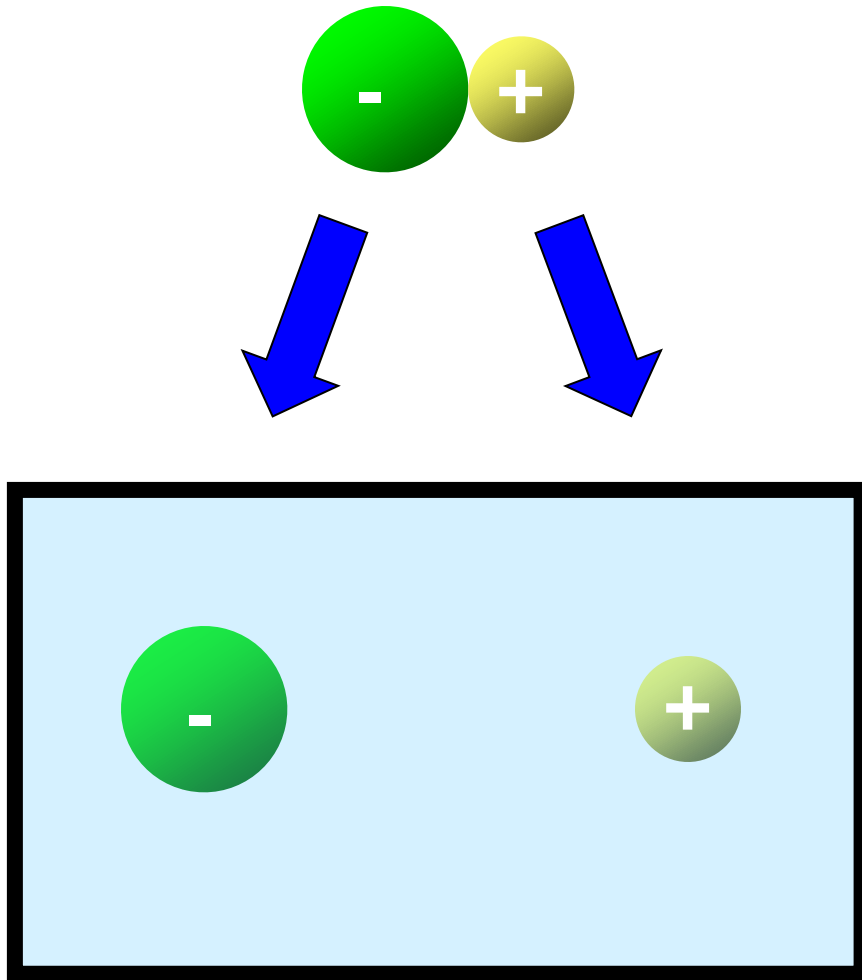
electrolyte

Solutions of Electrolytes, cont'd



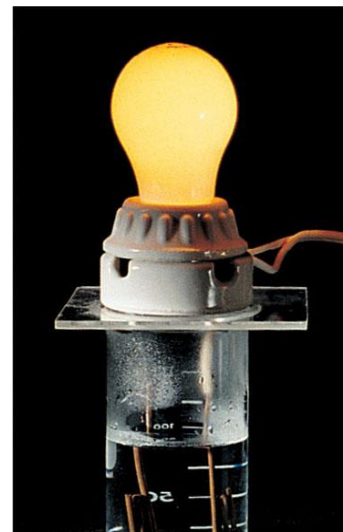
- One mole of propylene glycol gives one mole of dissolved solute particles.
- There are no charged species present
 - non-electrolyte

Solutions of Strong Electrolytes

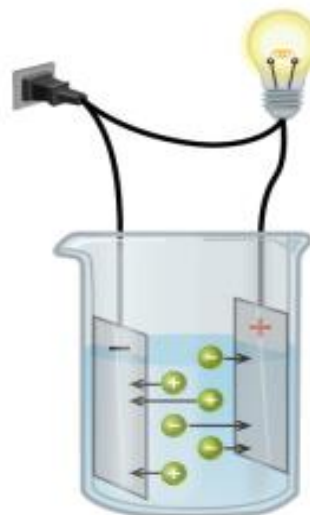


- One mole of NaCl gives **two** moles of dissolved solute particles.
 - **Electrolyte**
- Movement of the charged ions conducts electricity

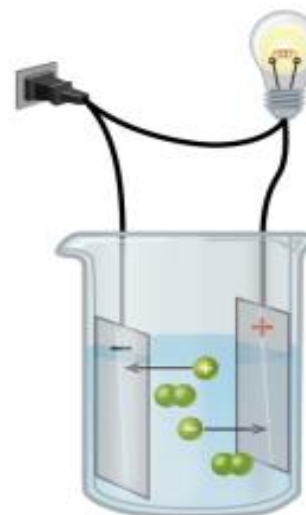
Conductivity of three types of electrolyte solutions



ethanol
No conductivity



KCl
High conductivity



acetic acid solution
Low conductivity

Figure 11.6 Solutions of nonelectrolytes such as ethanol do not contain dissolved ions and cannot conduct electricity. Solutions of electrolytes contain ions that permit the passage of electricity. The conductivity of an electrolyte solution is related to the strength of the electrolyte.

Solutions of Electrolytes, cont'd



- Electrolytes have anomalous colligative properties.
- Allowed Arrhenius to determine that they dissociate in solution.

Svante Arrhenius (1859-1927)

Water and other polar molecules are attracted to ions.

The electrostatic attraction between an ion and a molecule with a dipole is called an **ion-dipole attraction**.

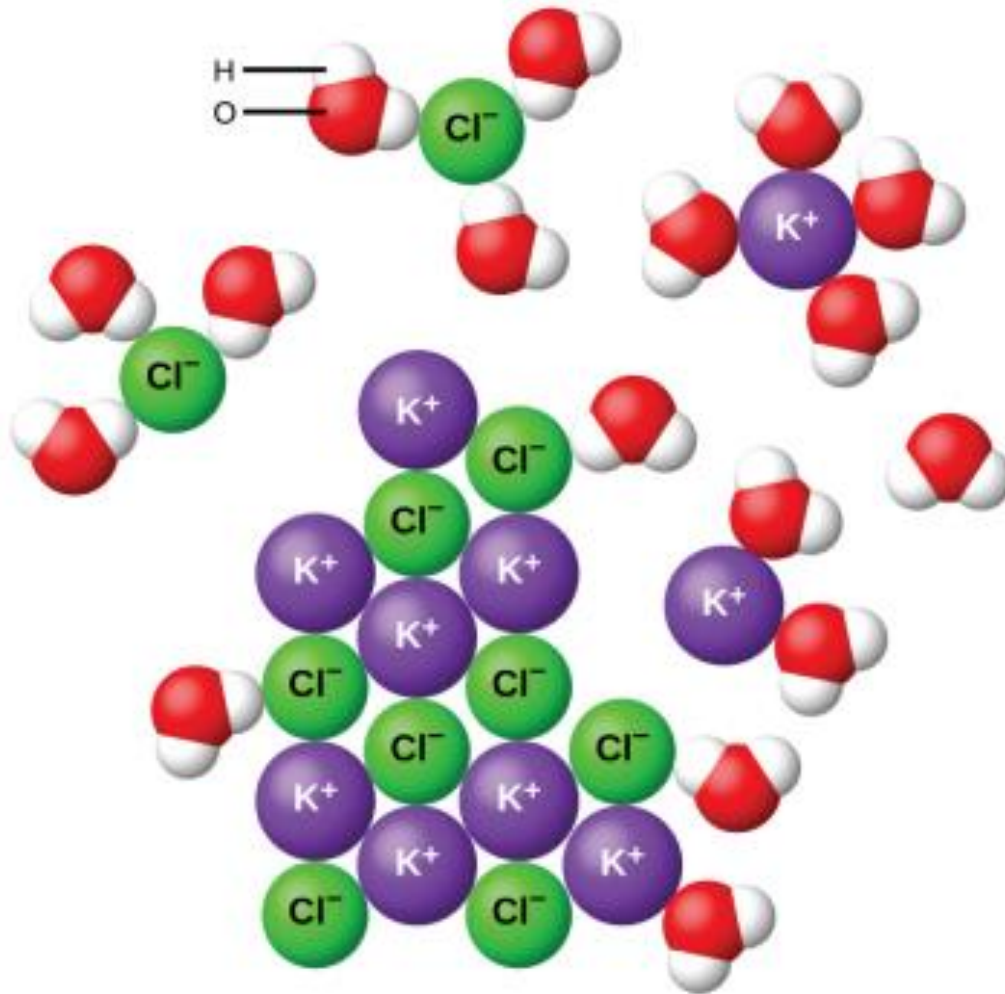
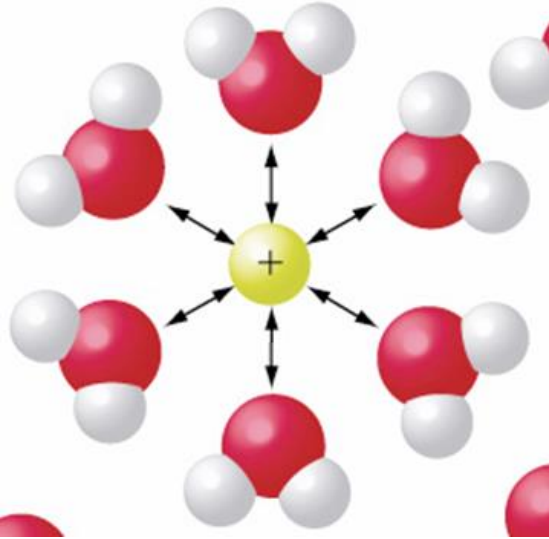
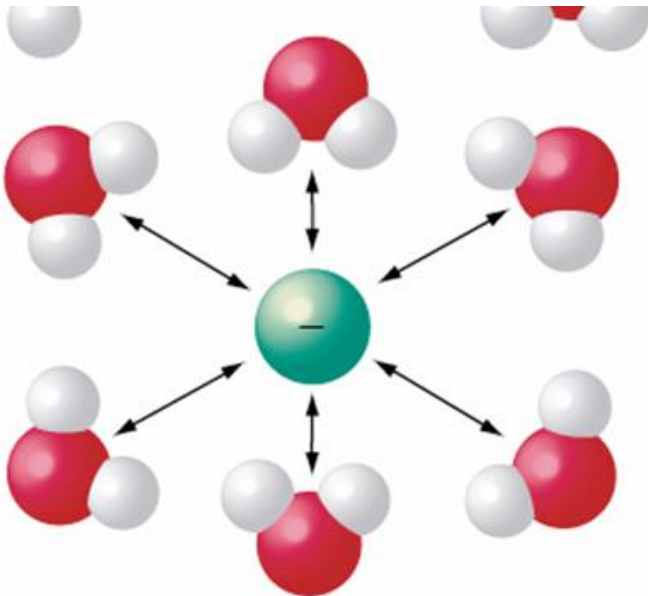


Figure 11.7 As potassium chloride (KCl) dissolves in water, the ions are hydrated. The polar water molecules are attracted by the charges on the K^+ and Cl^- ions. Water molecules in front of and behind the ions are not shown.

Formation of Ionic Solutions

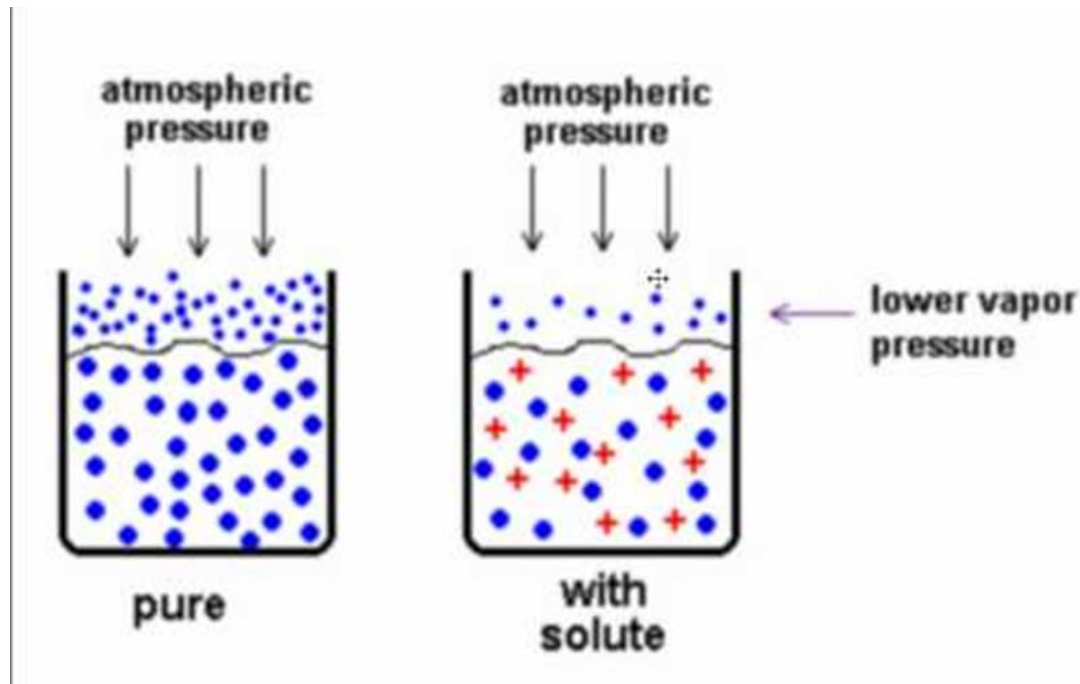


- Cations (+ve) preferentially interact with water oxygen atoms (negative partial charge)



- Anions (-ve) preferentially interact with water hydrogen atoms (positive partial charge)

Vapor Pressure of Solutions



- What is the vapor pressure above a solution?

Raoult's Law



- French chemist:
- Francois-Marie Raoult (1830-1901) discovered that in ideal cases the **partial pressure of a vapor above a solution depends on the mole fraction (x) of liquid in the solution.**

(Recall: mole fraction =
moles of solute $\div$ total moles of solution)

Raoult's Law, cont'd

$$P_A = x_A P_A^\circ$$

Partial
pressure
of **vapor** A

Mole
fraction of
liquid A

Vapor
pressure of
pure A

For ideal solutions, $\Delta H \sim 0$.
Also for dilute solutions.

Vapour Pressure Lowering

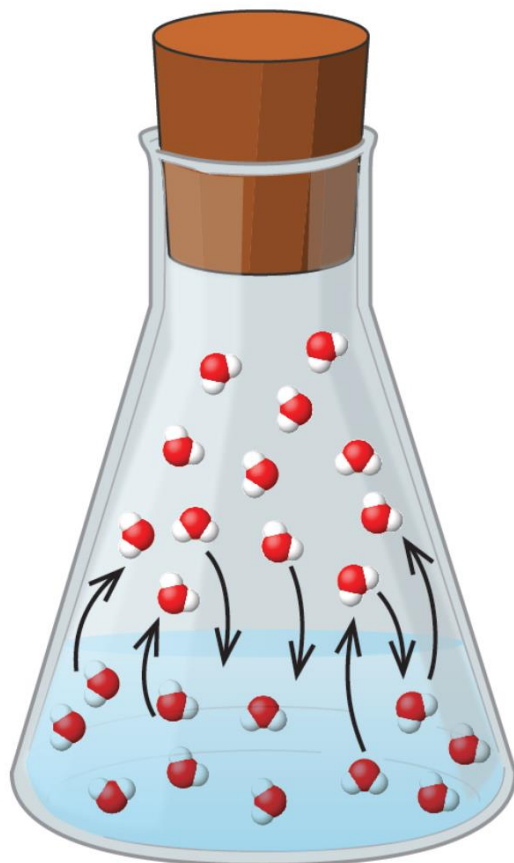
Raoult's Law:

$$P_{\text{solvent}} = X_{\text{solvent}} \times P^{\circ}_{\text{solvent}}$$

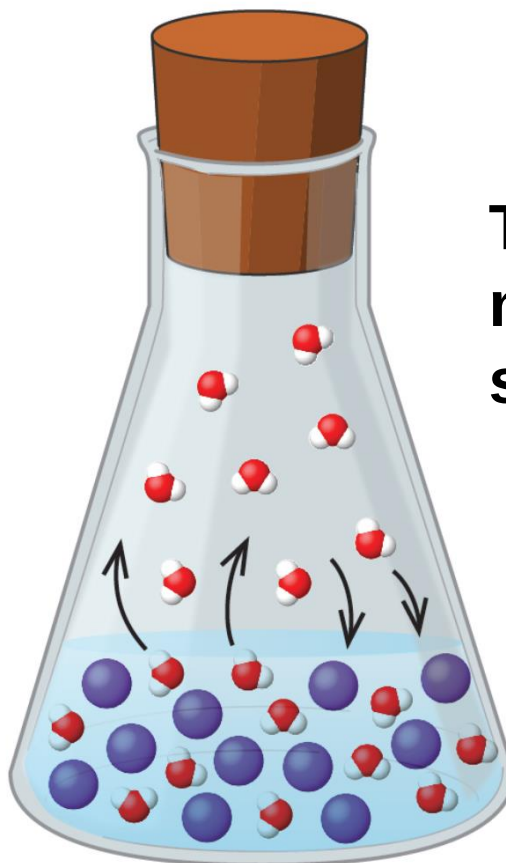
$$X_{\text{solvent}} = 1 - X_{\text{solute}}$$

$$\Delta P = X_{\text{solute}} \times P^{\circ}_{\text{solvent}}$$

$$\text{where } \Delta P = P^{\circ}_{\text{solvent}} - P_{\text{solvent}}$$



(a) Pure water



(b) Aqueous solution

**This is applied to
non-volatile
solute.**